Applied Econometrics: Session 10

A Basic Introduction to Macroeconometrics

PhD Intensive Course

Lecture 10

April 29, 2026

How This Lecture Fits the Course

The first part of the course focused on micro-style causal designs: experiments, controls, IV, matching, DiD, synthetic control, and RD.

Macroeconometrics asks related questions with different data:

- observations are usually indexed by time: $t = 1, \ldots, T$;
- outcomes are persistent and move together;
- one aggregate shock can affect many variables at once;
- sample sizes are often small: 60 quarterly observations is only 15 years.

Core message

In macroeconometrics, time dependence is not a nuisance. It is usually the object of interest.

Roadmap for a 2.5 Hour Introduction

Time	Topic
0–20 min	What macroeconometrics studies; why time-series data are special
20–55 min	Trends, growth rates, stationarity, autocorrelation, and unit roots
55–90 min	Univariate dynamics: AR models, persistence, and forecasts
90–125 min	Multivariate dynamics: VARs, Granger causality, and impulse responses
125–145 min	Identification: why a VAR residual is not automatically a structural shock
145–150 min	Checklist and takeaways

Learning Goals

By the end, students should be able to:

- distinguish forecasting, description, and causal/structural analysis;
- explain why serial correlation changes standard econometric reasoning;
- recognize trends, growth rates, stationarity, and unit-root behavior;
- estimate and interpret a simple autoregression;
- explain what a VAR is and what an impulse response measures;
- state why macro shocks require identifying assumptions.

What Is Macroeconometrics?

Macroeconometrics is econometrics for aggregate time-series and panel-time-series data.

- GDP, consumption, investment, inflation, unemployment, wages.
- Interest rates, credit, exchange rates, house prices, asset prices.
- Fiscal variables: spending, taxes, debt, deficits.
- Cross-country or regional macro panels.

Simple definition

Macroeconometrics studies how aggregate economic variables move over time, forecast each other, and respond to shocks and policy changes.

Three Different Tasks

The same dataset can be used for three different purposes.

Task	Question	Example
Description	What happened?	Inflation rose before unemployment fell.
Forecasting	What will happen next?	Predict next quarter's GDP growth.
Structural analysis	What caused what?	Estimate the effect of a monetary-policy shock.

Warning

A good forecasting model is not automatically a good causal model.

Why Macro Data Are Hard

Macro data are not just large versions of cross-sectional data.

- **Persistence:** high GDP today predicts high GDP tomorrow.
- **Common shocks:** recessions move many variables together.
- **Policy feedback:** central banks react to inflation and output.
- **Small samples:** annual and quarterly data give limited T.
- **Nonstationarity:** many levels trend over time.
- **Expectations:** decisions depend on anticipated future policy.

Running Example

We will use the same language for several macro variables:

$$y_t = \text{output growth}, \qquad \pi_t = \text{inflation}, \qquad i_t = \text{policy interest rate}.$$

Typical questions:

- Does inflation help forecast future interest rates?
- How persistent is output growth?
- What happens to output and inflation after a monetary tightening?
- Did two trending variables move together for economic reasons or just because both trended?

From Cross Sections to Time Series

In a cross section, observations are indexed by units:

$$i = 1, \ldots, n.$$

In a time series, observations are indexed by time:

$$t = 1, \ldots, T.$$

Key difference

y_t and y_{t-1} are usually related. Treating them as independent observations is often wrong.

Lag Notation

A lag is a past value of the same variable:

$$y_{t-1}, \quad y_{t-2}, \quad y_{t-3}.$$

The lag operator L is shorthand:

$$Ly_t = y_{t-1}, \qquad L^2 y_t = y_{t-2}.$$

Differences are written as

$$\Delta y_t = y_t - y_{t-1} = (1 - L)y_t.$$

Always Plot the Series First

Before estimating a model, plot the data.

- Is the variable a level, log level, growth rate, or change?
- Is there a trend?
- Are there breaks: crises, pandemics, wars, policy-regime changes?
- Are there outliers or measurement changes?
- Is the frequency annual, quarterly, monthly, or daily?

Practical rule

A macro regression that has not been preceded by a time-series plot is easy to misunderstand.

Levels, Logs, and Growth Rates

For a positive variable Y_t, define the log level:

$$y_t = \log(Y_t).$$

The log growth rate is

$$\Delta y_t = \log(Y_t) - \log(Y_{t-1}) \approx \frac{Y_t - Y_{t-1}}{Y_{t-1}}.$$

- GDP level is usually trending.
- GDP growth is often closer to stationary.
- Inflation is already a rate of price growth.

Stationarity

A time series is weakly stationary if:

$$\mathbb{E}[y_t] = \mu \quad \text{for all } t,$$

$$\mathrm{Var}(y_t) = \sigma_y^2 \quad \text{for all } t,$$

and

$$\mathrm{Cov}(y_t, y_{t-k})$$

depends on the lag k, not on the calendar date t.

Intuition

The process can be persistent, but its basic probabilistic behavior is stable over time.

Appendix details

Autocorrelation

The lag-k autocorrelation is

$$\rho_k = \mathrm{Corr}(y_t, y_{t-k}) = \frac{\mathrm{Cov}(y_t, y_{t-k})}{\mathrm{Var}(y_t)}.$$

- If ρ_1 is large, the series is persistent.
- If ρ_k decays slowly, shocks fade slowly.
- The autocorrelation function is a basic descriptive tool.

White Noise

A useful benchmark is white noise:

$$\varepsilon_t \sim (0, \sigma^2), \qquad \mathrm{Cov}(\varepsilon_t, \varepsilon_s) = 0 \quad \text{for } t \neq s.$$

- White noise is unpredictable from its own past.
- It can still have large variance.
- Many macro models describe observed variables as responses to shocks.

Do not confuse

White noise means no serial correlation. It does not mean economically unimportant.

The AR(1) Model

The simplest persistent time-series model is

$$y_t = c + \phi y_{t-1} + \varepsilon_t, \qquad \varepsilon_t \sim (0, \sigma^2).$$

- c controls the unconditional mean.
- ϕ measures persistence.
- ε_t is the one-period-ahead forecast error.

Stationarity condition

The AR(1) is stationary if $|\phi| < 1$.

Appendix details

Interpreting Persistence

Suppose

$$y_t = c + \phi y_{t-1} + \varepsilon_t.$$

If a shock raises y_t by 1 today, the expected effect is:

$$1, \quad \phi, \quad \phi^2, \quad \phi^3, \ldots$$

over future periods.

- $\phi = 0.2$: shock disappears quickly.
- $\phi = 0.8$: shock is persistent.
- $\phi = 1$: shock never dies out; this is a unit root.

Appendix details

Forecasting with an AR(1)

The one-step-ahead forecast is

$$\widehat{y}_{t+1|t} = \widehat{c} + \widehat{\phi} y_t.$$

The h-step-ahead forecast is

$$\widehat{y}_{t+h|t} = \widehat{\mu} + \widehat{\phi}^h (y_t - \widehat{\mu}), \qquad \widehat{\mu} = \frac{\widehat{c}}{1 - \widehat{\phi}}.$$

Interpretation

Forecasts revert to the mean if $|\widehat{\phi}| < 1$. They do not revert if $\widehat{\phi}$ is close to one.

Estimating the AR(1)

Estimate

$$y_t = c + \phi y_{t-1} + \varepsilon_t$$

by OLS on observations $t = 2, \ldots, T$.

- Dependent variable: y_t.
- Regressor: y_{t-1}.
- Coefficient: persistence.
- Residual: forecast error, not necessarily a structural shock.

Small-sample warning

With highly persistent data and small T, estimates and standard errors can be fragile.

Unit Roots

A unit-root process has

$$y_t = y_{t-1} + \varepsilon_t.$$

Then

$$y_t = y_0 + \sum_{s=1}^{t} \varepsilon_s.$$

- Shocks have permanent effects.
- The variance grows with time.
- Usual stationary regression intuition breaks down.

Spurious Regression

If two unrelated variables both trend or both have unit roots, the regression

$$y_t = \alpha + \beta x_t + u_t$$

can look impressive even when x_t has no economic effect on y_t.

Symptom

High R^2, significant t-statistics, and strongly autocorrelated residuals in a regression of unrelated trending series.

- Plot the data.
- Consider growth rates or first differences.
- Check whether there is a cointegrating relationship.

Appendix details

Cointegration

Two series can each be nonstationary, but a linear combination can be stationary:

$$y_t - \beta x_t \quad \text{is stationary.}$$

Intuition

The two variables may drift over time, but they do not drift arbitrarily far apart.

Examples where cointegration may be plausible:

- consumption and income;
- short and long interest rates;
- prices in closely connected markets;
- money, prices, and output under some monetary theories.

Why Multivariate Models?

Macro variables are jointly determined.

- Inflation affects monetary policy.
- Monetary policy affects output and inflation.
- Output affects unemployment, tax revenue, and credit.
- Financial variables react quickly to expectations.

Problem

A single-equation model can hide feedback among variables.

The VAR(1) Model

Let

$$z_t = \begin{bmatrix} y_t \\ \pi_t \\ i_t \end{bmatrix}.$$

A VAR(1) is

$$z_t = a + Az_{t-1} + u_t, \qquad \mathbb{E}[u_t] = 0, \quad \mathbb{E}[u_t u_t'] = \Sigma_u.$$

- Each variable is regressed on lagged values of all variables.
- The matrix A summarizes dynamic interactions.
- u_t is a vector of reduced-form forecast errors.

Appendix details

VAR Equations in Words

A three-variable VAR says:

$$\text{output today} = \text{lags of output, inflation, interest rates} + \text{forecast error.}$$

$$\text{inflation today} = \text{lags of output, inflation, interest rates} + \text{forecast error.}$$

$$\text{interest rate today} = \text{lags of output, inflation, interest rates} + \text{forecast error.}$$

Beginner intuition

A VAR is a disciplined way to let macro variables predict each other dynamically.

Choosing the Lag Length

A VAR(p) uses p lags:

$$z_t = a + A_1 z_{t-1} + \cdots + A_p z_{t-p} + u_t.$$

More lags can capture richer dynamics but cost degrees of freedom.

- Use economic judgment about timing.
- Check residual autocorrelation.
- Compare information criteria such as AIC and BIC.
- Remember that macro samples are often small.

Granger Causality

Variable x_t Granger-causes y_t if lags of x_t help forecast y_t after controlling for lags of y_t.
Example:

$$y_t = \alpha + \rho y_{t-1} + \gamma x_{t-1} + u_t.$$

Test

$$H_0 : \gamma = 0.$$

Important

Granger causality is predictive content. It is not automatically causal in the treatment-effect sense.

Impulse Responses

An impulse response traces what happens after a shock.

- Shock: an unexpected movement in one variable or structural disturbance.
- Response: predicted path of variables after the shock.
- Horizon: period $0, 1, 2, \ldots, H$.

For a VAR(1), if the initial shock is s, then

$$\mathrm{IRF}(h) = A^h s.$$

Appendix details

Reduced-Form Errors Are Not Structural Shocks

A VAR residual u_t is a forecast error:

$$z_t - \mathbb{E}[z_t \mid z_{t-1}, z_{t-2}, \ldots].$$

But a structural shock is an economically meaningful disturbance:

- monetary-policy shock;
- technology shock;
- demand shock;
- fiscal spending shock.

Identification problem

The data give reduced-form forecast errors. Turning them into structural shocks requires assumptions.

Common Identification Strategies

Macroeconomists identify shocks using restrictions or external information.

- **Recursive ordering:** some variables react within the period, others react with delay.
- **Sign restrictions:** a monetary tightening should raise rates and lower prices/output over some horizon.
- **External instruments:** use high-frequency surprises or narrative measures as shock proxies.
- **Natural experiments:** use institutional rules or policy discontinuities when available.

Connection to earlier lectures

This is the macro version of the same causal problem: what variation identifies the effect?

Local Projections

A popular alternative to VAR impulse responses is the local projection:

$$y_{t+h} = \alpha_h + \beta_h shock_t + \Gamma'_h controls_t + e_{t+h}.$$

Estimate a separate regression for each horizon h.

- Flexible and easy to explain.
- Works naturally with nonlinearities and interactions.
- Still requires an identified shock.
- Standard errors must handle serial correlation across horizons.

A Basic Macroeconometric Workflow

1. State the question: description, forecast, or structural effect.
2. Define variables, frequency, sample, transformations, and timing.
3. Plot all variables.
4. Decide levels, logs, growth rates, or differences.
5. Estimate a simple benchmark model.
6. Check residuals and robustness.
7. For causal claims, state the identifying assumptions.
8. Report magnitudes in economic units.

Python Example for This Lecture

The script

`lectures/code/lecture-10-macroeconometrics.py`

demonstrates three ideas:

- estimating an AR(1) and forecasting;
- seeing a spurious regression from unrelated random walks;
- estimating a two-variable VAR and computing impulse responses.

Student task

Change the persistence parameter and watch how forecasts and impulse responses change.

What to Report in a Basic Macro Exercise

Report enough for the reader to reconstruct the analysis:

- variables, definitions, units, and data frequency;
- sample start/end dates and missing-data treatment;
- transformations: log, difference, annualization, detrending;
- model specification and lag length;
- standard-error choice;
- diagnostics: residual autocorrelation, stability, outliers;
- identification assumptions for shocks;
- robustness to sample period and lag length.

Common Beginner Mistakes

- Regressing two trending variables in levels and treating the result as causal.
- Calling Granger causality "true causality."
- Forgetting that policy responds to the economy.
- Reading a VAR residual as a structural shock without identification.
- Using too many lags in a small quarterly sample.
- Ignoring breaks such as crises or major policy-regime changes.
- Reporting coefficients without translating them into economic units.

Takeaways

1. Macro data are dynamic: the past predicts the present.
2. Stationarity matters because many standard tools rely on stable distributions.
3. AR models teach persistence and forecasting.
4. VARs describe joint dynamics among macro variables.
5. Impulse responses are useful, but structural interpretation needs identification.
6. Always separate forecasting success from causal credibility.

Appendix A1: AR(1) Mean and Stationarity

Start with

$$y_t = c + \phi y_{t-1} + \varepsilon_t, \qquad \mathbb{E}[\varepsilon_t] = 0.$$

If the mean is constant, $\mathbb{E}[y_t] = \mathbb{E}[y_{t-1}] = \mu$. Then

$$\mu = c + \phi\mu \quad \Rightarrow \quad \mu = \frac{c}{1-\phi}.$$

Recursive substitution gives

$$y_t = \mu + \phi^j (y_{t-j} - \mu) + \sum_{\ell=0}^{j-1} \phi^\ell \varepsilon_{t-\ell}.$$

For the influence of the distant past to vanish, we need $|\phi| < 1$.

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Appendix A2: AR(1) Variance

If $|\phi| < 1$, write

$$y_t - \mu = \phi(y_{t-1} - \mu) + \varepsilon_t.$$

Taking variances and using $\mathrm{Cov}(y_{t-1}, \varepsilon_t) = 0$:

$$\mathrm{Var}(y_t) = \phi^2 \mathrm{Var}(y_{t-1}) + \sigma^2.$$

Stationarity means $\mathrm{Var}(y_t) = \mathrm{Var}(y_{t-1}) = \sigma_y^2$. Hence

$$\sigma_y^2 = \phi^2 \sigma_y^2 + \sigma^2 \quad \Rightarrow \quad \sigma_y^2 = \frac{\sigma^2}{1 - \phi^2}.$$

This is finite only if $|\phi| < 1$.

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Appendix A3: AR(1) Forecast Formula

With mean $\mu = c/(1-\phi)$,

$$y_t - \mu = \phi(y_{t-1} - \mu) + \varepsilon_t.$$

The one-step conditional expectation is

$$\mathbb{E}[y_{t+1} - \mu \mid y_t] = \phi(y_t - \mu).$$

Iterating forward:

$$\mathbb{E}[y_{t+h} \mid y_t] = \mu + \phi^h(y_t - \mu).$$

The same logic gives shock responses $1, \phi, \phi^2, \ldots$.

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Appendix A4: Why Unit Roots Create Trouble

If

$$y_t = y_{t-1} + \varepsilon_t,$$

then

$$y_t = y_0 + \sum_{s=1}^{t} \varepsilon_s.$$

The variance is

$$\mathrm{Var}(y_t) = t\sigma^2$$

when shocks are independent with variance σ^2.
Since the variance changes with t, the process is not stationary. Two independent random walks can appear related because both wander persistently.

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Appendix A5: VAR Companion Form

A VAR(p) is

$$z_t = a + A_1 z_{t-1} + \cdots + A_p z_{t-p} + u_t.$$

Stack current and lagged values:

$$Z_t = \begin{bmatrix} z_t \\ z_{t-1} \\ \vdots \\ z_{t-p+1} \end{bmatrix}.$$

Then

$$Z_t = \alpha + F Z_{t-1} + U_t.$$

This turns the VAR(p) into a larger VAR(1), which makes forecasting and impulse-response calculations easier.

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Appendix A6: VAR Impulse Response Recursion

For a VAR(1),

$$z_t = a + Az_{t-1} + u_t.$$

Suppose a shock changes z_t by vector s at horizon 0.

$$\mathrm{IRF}(0) = s.$$

One period later:

$$\mathrm{IRF}(1) = As.$$

Two periods later:

$$\mathrm{IRF}(2) = A^2 s.$$

Therefore

$$\mathrm{IRF}(h) = A^h s.$$

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Appendix A7: Cholesky Identification

Suppose reduced-form errors satisfy

$$\mathbb{E}[u_t u_t'] = \Sigma_u.$$

Cholesky identification chooses a lower-triangular matrix B such that

$$\Sigma_u = BB'.$$

Then

$$u_t = B\eta_t, \qquad \mathbb{E}[\eta_t \eta_t'] = I.$$

The first structural shock can affect all variables contemporaneously, the second cannot affect the first contemporaneously, and so on. This is an identifying assumption, not a fact delivered by the data.

Appendix A8: Serial Correlation and Standard Errors

In a simple regression

$$y_t = \alpha + \beta x_t + u_t,$$

usual OLS standard errors assume errors are not serially correlated.
If u_t is correlated with u_{t-1}, then the usual standard error can be too small.
Common responses:

- model the dynamics directly;
- use heteroskedasticity-and-autocorrelation consistent standard errors;
- aggregate or transform the data only when economically justified;
- be transparent about small-sample uncertainty.